Sean Luka Girgis

Senior Capacity Management / Infrastructure Analytics Engineer

214-315-2190 | seanlgirgis@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Senior Capacity Management and Infrastructure Analytics Engineer with 20+ years of enterprise IT experience across banking, performance engineering, observability, data pipelines, and executive reporting. Strong fit for capacity strategy, governance, KPI design, dashboarding, forecasting, risk reporting, Python automation, and cross-functional infrastructure/application/SRE partnership. At Citi, built capacity and telemetry pipelines for 6,000+ endpoints, integrated BMC TrueSight/TSCO, CA APM, and AppDynamics data, and delivered forecasting and executive insights for technology-capacity decisions.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Owned and improved enterprise capacity analytics workflows for banking infrastructure, translating CPU, utilization, telemetry, and platform-capacity signals into actionable risk reporting for senior stakeholders.
- Architected automated Python and Pandas ETL pipelines ingesting P95 telemetry from 6,000+ endpoints through BMC TrueSight/TSCO and related monitoring feeds, replacing manual reporting with governed repeatable workflows.
- Built capacity forecasting models with Prophet and scikit-learn to predict infrastructure bottlenecks months ahead and support proactive provisioning, headroom planning, and remediation decisions.
- Integrated BMC TrueSight/TSCO, CA Wily/APM, and AppDynamics data into unified Oracle reporting structures to support reconciled dashboards, trend analysis, and executive capacity views.
- Designed optimized Oracle schemas and historical retention models for capacity, telemetry, and seasonal-risk analysis across large-scale enterprise environments.
- Developed interactive Streamlit, matplotlib, seaborn, and plotly dashboards to surface real-time capacity insights, utilization trends, risk signals, and stakeholder-ready performance summaries.
- Supported governance and operational discipline through documentation, data quality checks, repeatable runbooks, issue analysis, and structured reporting aligned to production-support expectations.
- Partnered with application, infrastructure, and operations teams to identify underutilized assets, capacity constraints, and performance risks, helping drive consolidation and cost-aware infrastructure decisions.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, strengthening visibility into production application performance and customer-impacting behavior.
- Led the FAST project, mining real-user and synthetic performance data to identify optimization opportunities and convert telemetry into engineering recommendations.
- Implemented End-User Monitoring and end-to-end transaction tracing across web and mobile systems to improve application observability and troubleshooting precision.

- Integrated HP Performance Center with Dynatrace to correlate load-test behavior with APM telemetry and support performance-based release decisions.
- Presented performance findings, trend analysis, and actionable optimization guidance to engineering and operations stakeholders.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for an enterprise financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules that improved application visibility and became reusable standards across client environments.
- Led CA APM upgrade and standardization work, coordinating cross-functional teams while maintaining production monitoring continuity.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and reduce manual weekly export effort.
- Collaborated with J2EE/WebLogic performance teams to diagnose application bottlenecks using APM telemetry, SQL/Oracle data, thread behavior, and infrastructure signals.
- Delivered documentation, onboarding support, and technical training for monitoring standards, agent deployments, and production troubleshooting practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits and resource bottlenecks across JDBC connections, threads, heap memory, CPU, and garbage collection behavior.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to provide real-time visibility during load and performance testing cycles.
- Documented performance baselines and regression thresholds used by engineering teams for release-gate decisions and production-readiness discussions.
- Partnered with development and infrastructure teams to interpret load-test results and recommend tuning actions across application and platform layers.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict reliability and performance expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness across large transaction datasets during phased cutover.
- Automated schema conversion and data-validation workflows to protect data integrity across relational database migration paths.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability telecom billing interfaces using C, C++, Pro*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and production Unix environments.
- Improved throughput and database performance in billing processes through SQL tuning, sequence optimization, memory reduction, and interface-level performance analysis.
- Built and maintained reporting, invoicing, settlement, and integration components across full billing lifecycle systems for large-scale telecom operations.

- Automated WebLogic/WebSphere administration and deployment support with Korn Shell scripting, reducing manual operational effort.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

HorizonScale — AI Capacity Forecasting Engine
Designed an AI-driven capacity forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform enterprise telemetry into bottleneck predictions and operational decision support.
Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Telemetry Analytics, Capacity Forecasting

Enterprise Capacity Telemetry Reporting Platform
Built automated capacity reporting workflows that combined infrastructure telemetry, APM signals, Oracle data models, and stakeholder dashboards to support enterprise capacity planning and risk review.
Technologies: Python, Pandas, Oracle, SQL, BMC TrueSight/TSCO, CA APM, AppDynamics, Dashboards

FAST Project
Mined real-user and synthetic performance metrics from Dynatrace and Gomez Synthetic Monitoring to identify optimization opportunities and provide engineering teams with actionable performance recommendations.
Technologies: Dynatrace, Gomez Synthetic Monitoring, Telemetry Analytics, Performance Engineering

Serverless Lakehouse Platform (AWS)
Designed a serverless data platform pattern using S3, Glue, Athena, Redshift, Bedrock, PySpark, Parquet, and Snappy compression for cloud-scale ingestion, transformation, and query use cases.
Technologies: AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet